

# DHANUSH GOWDHAMAN

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## Education

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- **University of Pennsylvania, Philadelphia, PA** *Aug 2025 – May 2027 (Expected)*  
MSE, Computer and Information Science **CGPA: 3.53/4.0**
- **PSG College of Technology, Coimbatore, India** *Aug 2020 – May 2024*  
BE, Computer Science and Engineering **CGPA: 3.93/4.0**

## Work Experience

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- **Course Development Assistant, AI Capstone - Penn Engineering Online** *Oct 2025 – Present*
  - Develop graduate-level curriculum — lecture slides, notes, and hands-on Jupyter notebooks — for a new project-based AI Capstone course (led by Prof. Chris Callison-Burch) in Penn’s MSE in AI Online program, spanning LLM application patterns, RAG, agentic orchestration, fine-tuning (LoRA/PEFT), evaluation harness design, and containerized deployment.
  - Built reference implementations and scaffolded starter code, including a full-stack AI application delivered as staged milestones for students to extend and complete.
- **Software Developer, IBM India Pvt Ltd** *Jul 2024 – Jul 2025*
  - Shipped an internal VS Code extension (LSP support, document parsing, ref-def navigation) with Generative AI-based explainability for legacy languages — adopted by 50 developers within 3 months.
  - Migrated builds from a single static machine to a containerized CI/CD pipeline (Docker), eliminating a single point of failure and enabling multiple parallel builds across the team.
- **Software Developer Intern, IBM India Pvt Ltd** *Jan 2024 – Jun 2024*
  - Deployed CICS TX on Cloud using Podman, Kubernetes, OpenShift, and AWS EKS, reducing maintenance overhead and saving 2 person-days of effort per quarter.

## Projects

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- **Voice DevOps Tool** (*1st Place — Cactus x DeepMind Hackathon, Washington D.C. 2026*) — Built a voice-driven DevOps assistant with a complexity-routing algorithm that dynamically dispatches simple commands to a local model (FunctionGemma) and complex tasks to the cloud (Gemini), optimizing for latency and cost.
- **Flash-Decks (AI Flashcards Web App)** — Built a full-stack spaced-repetition study platform (React/TypeScript, FastAPI, PostgreSQL) with exam mode, SM-2 scheduling, and progress analytics; integrated LLM-powered deck generation and open-ended answer grading. Shipped with a GitHub Actions CI pipeline and 150+ automated tests at 93% backend coverage. [Flashcards-app](#)
- **ASL Interpreter & Translator** — Built a bidirectional ASL/English system: a real-time recognizer (TensorFlow, OpenCV, MediaPipe) classifying 24 static hand signs from webcam input at 99% accuracy (benchmarked across 7 models; published at IEEE ICCCNT 2024), plus an English-to-ASL pipeline using a locally-hosted Llama 3.1 (LM Studio) to simplify sentences into core ASL tokens and retrieve matching sign videos via similarity search. [ASL-Translator](#)
- **Cybersecurity CTF** — Placed 1st (2022) and 2nd (2021) in PSG CSEA CTF competitions; later organized 3 events and authored 25+ Linux, cryptography, DBMS, and Python challenges for 100+ participants. [Iris CTF](#), [Aerovision CTF](#)
- **Brand Prediction & Analysis** — Built a full-stack app (React, Express) for batch topic-brand text classification; the Node backend invokes a Python ML service (Multinomial Naive Bayes) that ingests an uploaded Excel sheet and returns it with an appended predictions column as a downloadable file. Containerized with Docker.

## Technical Skills

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**Languages:** Python, C/C++, Java, JavaScript/TypeScript, SQL

**Frameworks & Libraries:** React, Node.js/Express, FastAPI, Flask, TensorFlow, scikit-learn, OpenCV, MediaPipe

**Tools & Technologies:** Docker, Kubernetes, Podman, Git, CI/CD (GitHub Actions), PostgreSQL, REST APIs, AWS (Cloud Practitioner), Linux, OpenShift, Tailwind CSS, LM Studio

**AI/ML:** LLMs, RAG, NLP, Computer Vision

## Publications

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- “American Sign Language Alphabet Recognition using Teachable Machine”, The 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2024.  
DOI: [10.1109/ICCCNT61001.2024.10725877](https://doi.org/10.1109/ICCCNT61001.2024.10725877)